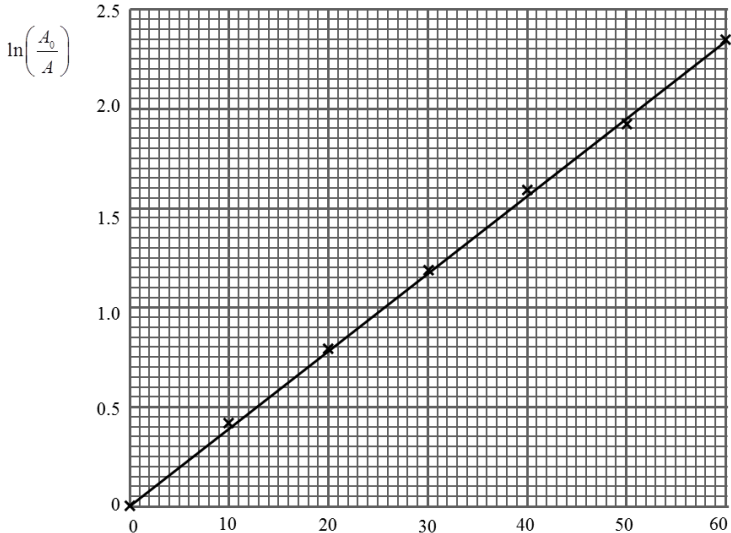


Question			Marking details	Marks available				Maths	Prac																								
				AO1	AO2	AO3	Total																										
6	(a)	(i)	$kx = mg$ (1) $x = \frac{0.150 \times 9.81}{7.5} = 0.196 \text{ m}$ answer (1)		2		2	2	2																								
		(ii)	$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.150}{7.5}}$ substitution (1) $= 0.889 \text{ s}$ (1) NB. No credit for use of $T = 2\pi\sqrt{\frac{l}{g}}$	1	1		2	2	2																								
	(b)		Drop in amplitude at beginning and end compared, e.g. 0.033 m and 0.005 m over first and last intervals. [Or rates of decrease: 0.0033 m per oscillation and 0.0005 m per oscillation] consider rate of decrease (1) Justify by noting the magnitude is larger at the start than at end of experiment. (1)			2	2		2																								
	(c)	(i)	When $n = 0$, $A = A_0 e^{\frac{0}{N}} = A_0 e^0 = A_0$ [Alternative: say $e^0 = 1$, so $A = A_0$.]			1	1	1	1																								
		(ii)	$A = A_0 e^{\frac{-n}{N}}$; $0.029 = 0.095 e^{\frac{-30}{N}}$ substitution (1) $N = \frac{-30}{\ln\left(\frac{0.029}{0.095}\right)}$ or equiv $\rightarrow N = 25.28 \cong 25$ answer (1)	1	1		2	2	2																								
	(d)	(i)	<table border="1"><thead><tr><th>Oscillation number (n)</th><th>Amplitude (A) /m</th><th></th></tr></thead><tbody><tr><td>0</td><td>0.095</td><td>0</td></tr><tr><td>10</td><td>0.062</td><td>0.43</td></tr><tr><td>20</td><td>0.043</td><td>0.79</td></tr><tr><td>30</td><td>0.029</td><td>1.19</td></tr><tr><td>40</td><td>0.019</td><td>1.61</td></tr><tr><td>50</td><td>0.014</td><td>1.91</td></tr><tr><td>60</td><td>0.009</td><td>2.36</td></tr></tbody></table> All three values correct (1) [accept 1.6, 1.9, 2.4 if consistent]	Oscillation number (n)	Amplitude (A) /m		0	0.095	0	10	0.062	0.43	20	0.043	0.79	30	0.029	1.19	40	0.019	1.61	50	0.014	1.91	60	0.009	2.36		1		1	1	1
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		(ii)	 Labelled axes with a suitable choice of scales (1) Oscillation number All points plotted correctly including origin (1) (ecf) Appropriate line of best fit [not necessarily through origin] ecf (1)	1						
		(iii)	Any two of: 1) It is a straight line, or positive gradient (1) 2) Passes through the origin (1) 3) The measured points lie essentially on a straight line or little scatter(1)			2	2		3	3
		(iv)	Gradient = $\frac{2.35 - 0}{60 - 0}$ (from graph) = 0.039 (ecf) (1) $N = \frac{1}{\text{gradient}}$ (1) = 25.6 (1) Accept answers in range 25 – 26. Use of a point on the line – allow. Use of a data point not on the line → 1st mark not available.			3	3		3	3
		(v)	Expected to be the one from (d)(iv) as straight line based on all points [The value in (c)(ii) is calculated from only a single point, which may be anomalous.]			1	1			1
			Question 6 total	3	7	9	19		14	19